

Explainable Common Spatial Pattern to Improve the Design of Brain-Computer Interfaces

Sofien Gannouni¹, Kais Belwafi^{2,1}

¹ Department of Computer Science, College of Computer and Information Sciences, King Saud University, Riyadh, KSA

² Department of Computer Engineering, College of Computing and Informatics, University of Sharjah, UAE

Abstract—Common Spatial Pattern (CSP) is a widely utilized technique in ElectroEncephaloGraphy (EEG)-based Motor Imagery (MI) Brain-Computer Interfaces (BCIs) for feature extraction, owing to its simplicity and effectiveness in improving classification performance. However, traditional CSP methods have limitations, including sensitivity to outliers and prone to overfitting, random variations in accuracy, and a lack of interpretability due to the predefined number of features. This paper introduces a novel framework integrating CSP with Explainable Artificial Intelligence (XAI) techniques to address these limitations. The proposed explainable CSP algorithm enhances classification accuracy while providing insights into feature selection and spatial patterns contributing to predictive accuracy. Key contributions include a mechanism for explainable feature selection, dynamic optimization of the number of features guided by explainability metrics, and validation across different subjects. Results showed that the proposed XAI-CSP method can outperform traditional CSP by 3% and similar CSP variants, such as Filter Bank CSP (FBCSP), by nearly 8% in average classification accuracy.

Index Terms—Common Spatial Pattern (CSP), Brain-Computer Interface (BCI), ElectroEncephaloGram (EEG), Explainable AI

I. INTRODUCTION

The Common Spatial Pattern (CSP) technique is a widely used method in ElectroEncephaloGraphy (EEG)-based Motor Imagery (MI) Brain-Computer Interfaces (BCIs) for feature extraction. CSP is instrumental in enhancing the classification performance of MI tasks by optimizing the spatial filtering of EEG signals, which are often contaminated by artifacts and noise [1], [2]. The origins of these artifacts are both physiological, related to eye movements and muscle activation, and non-physiological, linked to environmental factors such as electrode placement and acquisition system [3]. These contaminants significantly impair the classification performance of motor imagery-based brain-computer interfaces.

In the literature, various advancements and applications of CSP in MI BCIs have been proposed, emphasizing its robustness, adaptability, and integration with other methods. In [1], Lee and al proposed a Generalized Lp-norm based CSP (GLCSP) (with $p < 1$) to enhance the traditional CSP algorithm based on L2-norm and L1-norm, which are very sensitive to outliers. The method utilized Lp-norm dispersion

of filtered EEG samples.

A recent study introduced the Variance Characteristic Preserving CSP (VPCSP) method [4]. It modifies the traditional CSP algorithm by incorporating a regularization item based on graph theory. It allows for the preservation of local variance characteristics while addressing the loss of abnormalities in the projected data.

CSP has been combined with advanced signal processing techniques, such as Power Spectrum Density (PSD), Discrete Wavelet Transform (DWT), Weighted-OverLapp-Add (WOLA) [5], iterative filtering, and filter bank methods, to enhance the accuracy and reliability of BCIs [6].

Despite the promise of CSP techniques in improving MI-BCI systems, the main remaining challenge is their adaptability to different subjects. Traditional CSP-based methods suffer from random variations in accuracy and a fixed number of features, which can lead to suboptimal classification performance. These limitations highlight the need for more robust and flexible approaches. The exploration of various CSP methods, including Riemannian CSP, explainable CSP, and deep learning-based CSP, continues to be crucial for advancing the field and enhancing the reliability and accuracy of MI-based BCIs. To address these issues, this paper proposes a new explainable CSP algorithm that not only improves classification accuracy but also provides a transparent framework to justify the selection of one feature over another, ensuring greater trust and interpretability in BCI systems.

A. Motivation & Contributions

While the CSP algorithm is widely used for feature extraction in EEG-based BCIs due to its simplicity and effectiveness, it often lacks interpretability. This limitation makes it difficult for practitioners to understand why specific features are selected or how they contribute to classification performance, particularly when adapting to new subjects or tasks. Integrating Explainable Artificial Intelligence (XAI) with CSP can address this issue, providing transparency and interpretability. This integration is crucial for improving trust in BCI systems, optimizing feature selection, and enabling highly accurate functional restitution or clinicians to make informed decisions based on explainable outputs. This work introduces a novel framework that integrates CSP with XAI techniques to enhance the interpretability of feature extraction in BCIs. The contributions include:

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Corresponding author: kbelwafi@sharjah.ac.ae

- 1) **Explainable Feature Selection:** A mechanism that identifies and explains the importance of spatial patterns in EEG signals, offering insights into the features contributing to classification accuracy.
- 2) **Dynamic Feature Optimization:** A method to determine the optimal number of features dynamically, guided by explainability metrics, to enhance classification performance.
- 3) **Validation Across Subjects:** Extensive experiments on public datasets demonstrate the framework's ability to maintain high classification accuracy while providing interpretable insights.

This integration paves the way for more transparent, adaptable, and user-trustworthy BCI systems.

B. Terminology and annotations

- A subject (s) refers to a person from whom a set of trials (t) is collected using a set of electrodes denoted by E .
- S_1 contains recording of set of subjects s , and $\mathfrak{t} \in \{1, 2\}$; where S_1 and S_2 corresponds to training and testing data, respectively.
- Each trial (t) belongs to left-hand η_1 or right-hand η_2 .
- The EEG signal are sampled with sampling frequency (F_s).
- T is a trial, with a size (X, E) , where $X = E_w F_s$ and E_w is the time window expressed in second.

C. Paper organization

The rest of the paper is organized as follows: The proposed framework is presented in Section II. The results are discussed in Sections III, and finally, the paper is concluded in Section IV.

II. PROPOSED METHODOLOGY

This section presents the experimental setup for the proposed framework. The framework consists of different blocks starting with the filtering block to remove artifacts, feature extraction to reduce data size, feature explanation to keep the pertinent information, and classification to associate each trial (t) with its corresponding label. Figure 1 presents a general overview of the framework.

A. Experimental setup

1) *Description of the datasets:* A public dataset is used in this study to analyse and validate the proposed approach. The datasets are provided by the BCI-competition containing in total the EEG recording of 9 subjects (s), where each one has two datasets (S_1). The brain activities are recorded using 22 electrodes placed on the scalp and sampled at a frequency rate of 256 Hz. The selected dataset contains EEG recordings of different subjects thinking to move left-hand, right-hand, both foot and tongue. We limited our study to classes related to left- and right-hand movements, as the CSP algorithm can only be applied to two classes.

2) *Software and tools:* The architecture of the AI-BCI framework is primarily developed using Python and Matlab environments. The features of the different datasets are extracted using the various versions of Regularized Common Spatial Pattern (RCSP) after removing the unwanted signals outside the α and β bands [7]. These bands contain the usefully SensoriMotor Rhythms (SMRs) [8]. The extracted features are then sent to Google Colaboratory (Colab) for processing using Graphical Processing Unit (GPU) to explain and identify the useful weighting vector before training the model. classifying new trials, and evaluating the framework.

B. Artifact removals

EEG signal processing begins by removing physiological and non-physiological artifacts using temporal filters, frequency filters, or adaptive filters. Temporal filtering based on Infinite Impulse Response (IIR) is applied to each trial ($t \in T$) to eliminate time-based artifacts, and it can be mathematically represented as (Eq 1):

$$y[n] = \sum_{k=0}^{M-1} b_k \cdot t([n-k], e) - \sum_{j=1}^{N-1} a_j \cdot y[n-j] \quad (1)$$

where $t([n-k], e)$ is the signal in electrode (e), $y[n]$ is the filtered signal, a_k and b_k are the filter coefficients with length M to keep spectrum in α and β bands. This process helps in enhancing the signal quality for both datasets (S_1 and S_2).

C. Feature extraction

The CSP is widely used for feature extraction of MI-EEG signals [9], [10]. It permits the determination of the spatial filter denoted by ($\mathcal{W}$) by maximizing the variance of one class while minimizing the other one. From this point of view, it is applicable in the two-class classification problem and it can be used in extracting features corresponding to more than two classes by using the One-Versus rest approach [11]. The objective function of the CSP is defined by:

$$J(\mathcal{W}) = \frac{\mathcal{W}^T C_1 \mathcal{W}}{\mathcal{W}^T C_2 \mathcal{W}} \quad (2)$$

where $\mathcal{W}$ is the spatial filter, C_1 and C_2 are respectively the covariance matrices of trials ($t \in \eta_1$) and ($t \in \eta_2$). The size of the spatial filter ($\mathcal{W}$) is $E \times E$, which is directly related to the number of electrodes (E) used in the recording process. Algorithm 1 resumes the different steps to estimate the spatial filter $\mathcal{W}$.

The existing research studies reduced the size of $\mathcal{W}$ to a fixed number ($\mathcal{N}$) (e.g., 3 or 4) of features corresponding to the maximum and minimum values [7], [12]. The question remains about the definition of the optimal value ($\mathcal{N}$) while the spatial filter ($\mathcal{W}$) tracks effectively the Event-Related Desynchronization/Synchronization (ERD/ERS) phenomenon. To address this issue, we suggest the adaptation of the RuleFit algorithm to choose the pertinent features. It was designed by Friedman and Popescu to learn sparse linear forms containing the interaction effects in the form of decision-making rules [13].

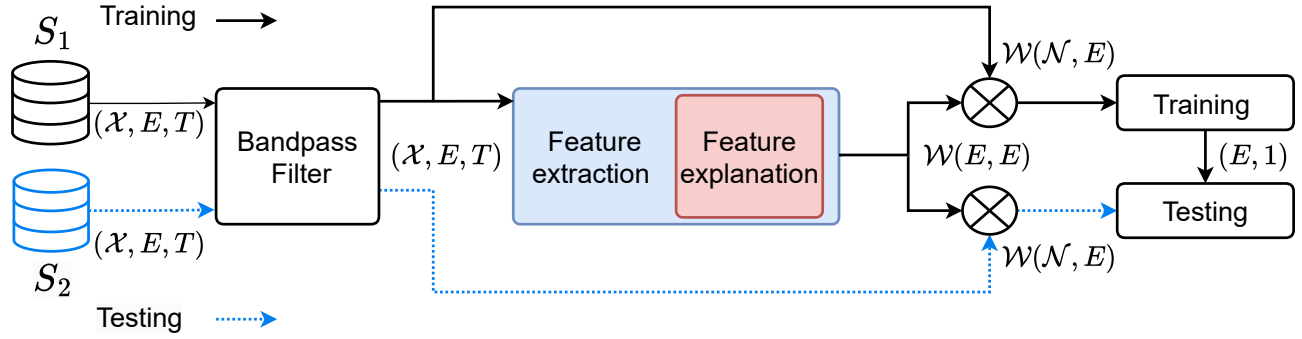


Fig. 1: General overview of the proposed framework

D. Feature explanation

The explainable CSP will be applied by analyzing the computed spatial filter ($\mathcal{W}$), computed by the CSP, and selecting the pertinent feature vectors. Our goal is to teach a sparse linear model that incorporates various features ($\mathcal{W}$) along with new features capturing the interactions between these ($\mathcal{W}$) features. RuleFit automatically generates rules from Decision Trees (DTs), where each path in the DT represents a decision rule. The optimal value ($\mathcal{N}$) corresponding to the size of ($\mathcal{W}$) is estimated at the end of the block by choosing the index of the useful features with the high importance score.

RuleFit comprises two key components:

- 1) decision rules derived from decision trees,
- 2) the fitting of a sparse linear model that incorporates both the original features and the newly generated rules.

To ensure robustness, original features are winsorized ($I_j(y_j)$) to limit the impact of outliers and normalized before combining with rules in the final model.

The model then calculates the total importance score for each feature ($IF_j(y)$), considering both the original features and their contributions within the decision rules. This feature attribution helps make the selection of important features transparent, allowing the model to explain its predictions effectively based on the relevance of each spatial pattern.

E. Classification

In our classification approach, we utilize Linear Discriminant Analysis (LDA) as a classical machine learning algorithm to benchmark performance and demonstrate the efficacy of the proposed explainable CSP method. This approach enables the preservation of relevant features while integrating traditional machine learning algorithms.

LDA is commonly used in BCI to classify brain signals. It is used to identify a linear combination of features that maximizes the separation between distinct types of signals, which is required for reliable data detection and classification [14].

Algorithm 1: Explainable CSP

Data: Sequence of $\mathcal{M}$ DTs
Data: Sequence of trials ($t \in \{\eta_1, \eta_2\}$)

- 1 $i \leftarrow 1, 2, \dots, \mathcal{M}$
- 2 $\mathbf{C}_1 = \frac{1}{\text{card}(\eta_1)} \mathbf{t}_{\in \eta_1} \mathbf{t}_{\in \eta_1}^T$
- 3 $\mathbf{C}_2 = \frac{1}{\text{card}(\eta_2)} \mathbf{t}_{\in \eta_2} \mathbf{X}_{\in \eta_2}^T$
- 4 $\mathbf{C} = \frac{1}{2} (\mathbf{C}_1 + \mathbf{C}_2) = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$
- 5 $\mathcal{W} = \mathbf{U}^T \mathbf{\Lambda}^{-1/2} \mathbf{U}$
- 6 $\mathcal{F}(x) = \hat{p}_0 + \sum_{i=1}^{\mathcal{M}} \hat{p}_i \mathcal{F}_i(\mathcal{W})$; where $\hat{p}_i$ are the weights and $\mathcal{F}_i(\mathcal{W})$ is the prediction function of the i^{th} DT
- 7 **foreach** $j \in T_i$ **do**
- 8 $r_i(\mathcal{W}) = \prod_{j \in T_i} I(y_j \in \zeta_{ji})$; where T_i is the set of features used in the i^{th} decision tree, and ζ_{ji} is an interval in the range of values of the features. The indicator function $I(y_j \in \zeta_{ji})$ is 1 when y_j is within this interval and 0 otherwise.
- 9 $l_j(y_j) = \min(\gamma_j^+, \max(\gamma_j^-, y_j))$; where γ_j^+ and γ_j^- are the quantiles of the distribution of the feature y_j .
- 10 $l_j^*(y_j) = 0.4 \cdot l_j(y_j) / \text{std}(l_j(y_j))$
- 11 $F(x) = \lambda_0 + \sum_{k=1}^K \phi_k r_k(y) + \sum_{j=1}^J \lambda_j l_j^*(y_j)$; where λ is the estimated weight for $\mathcal{W}$ and ϕ is for the generated rules.
- 12 $IF_j(y) = l_j(y) + \sum_{y_j \in r_k} \frac{l_k(x)}{m_k}$
- 13 $IF_j(Y) = \sum_{j=1}^J IF_j(y_j)$

F. Evaluation Metrics

An evaluation of the proposed system is performed to estimate the Model Performance Metrics using accuracy ($\mathcal{A}$ (Eq. 3)), precision ($\mathcal{P}$ (Eq. 4)), recall ($\mathcal{R}$ (Eq. 5)), and F1 score ($\mathcal{F}_1$ (Eq. 6)). These measurements are based on the standard confusion matrix parameters: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) [7].

$$\mathcal{A} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\mathcal{P} = \frac{TP}{TP + FP} \quad (4)$$

$$\mathcal{R} = \frac{TP}{TP + FN}; \quad (5)$$

$$\mathcal{F}_1 = \frac{2 \times \mathcal{P} \times \mathcal{R}}{\mathcal{P} + \mathcal{R}} \quad (6)$$

III. RESULTS AND DISCUSSION

Figure 2 presents the spatial filter $\mathcal{W}$ computed at the output of the CSP, sorted based on the proposed algorithm 1. $f_{i \in \{1 \dots \text{card}(E)\}}$ represents the label of the features. Four subjects were selected for analysis: two of them (S_7 and S_8) have shown very high performance in the literature, while the other two (S_2 and S_5) have demonstrated very low accuracy.

In the literature, the important features are selected statically, where the majority are selected the first and last three features ($f_{1 \dots 3} \cup f_{\text{card}(E) - 3 \dots \text{card}(E)}$). These features are assumed to contain useful information. Figure 2g shows clearly the important features are the set $\{f_1, f_2, f_3, f_{16}, f_{20}, f_{18}\}$ but when using the static selection two important features are discarded while taking two new less important ones. Similarly, for subject s_5 as depicted in Figure 2i, two very important features are ignored. Regarding the features of s_7 and s_8 (Figures ?? and ??), the main features are selected correctly even if one of these features is ignored. From this viewpoint, the static selection of the features should be reconsidered as it will work arbitrarily for some subjects and others not.

Table I presents the performance metrics of the proposed method, evaluated across 9 subjects. The results show significant variability across subjects, with subject 3 achieving the highest $\mathcal{A}$ (96.53%), $\mathcal{P}$ (100%), and $\mathcal{F}_1$ (96.40%), indicating near-perfect performance for this subject. On the other hand, subject 5 demonstrates the lowest performance, with an $\mathcal{A}$ of 61.11%, $\mathcal{P}$ of 76.67%, and $\mathcal{R}$ of 31.94%, resulting in a considerably lower $\mathcal{F}_1$ of 45.10%. It is worth noting that the performance of this subject tends to be consistently low in many studies, possibly due to incorrect labeling of trials or the subject being left-handed, which could affect the system's ability to classify their data accurately.

When averaged across all subjects, the proposed method achieves an overall $\mathcal{A}$ of 81%, $\mathcal{P}$ of 83.52%, and $\mathcal{R}$ of 75.35%. The $\mathcal{F}_1$, which balances precision and recall, stands at 78.93%. These results demonstrate good overall performance, with the system showing a strong ability to correctly classify positive instances, although subject-specific performance variation suggests the potential for further optimization.

The proposed algorithm demonstrates superior performance in terms of accuracy and explainability compared to traditional CSP methods like GLCSP, VPCSP, Weighted Tikhonov Regularized CSP (WTRCSP), and WOLA-CSP. The dynamic feature selection process ensures that the number of features is optimized for each subject, leading to consistent improvements in classification accuracy.

Table II presents a comparison of different variants of the CSP method, highlighting their reported accuracy, dataset usage, and adaptability across subjects. The dataset column indicates whether the reported accuracy is based on the same

TABLE I: Performance Metrics Across 9 Subjects

s	LDA			
	$\mathcal{A}(\%)$	$\mathcal{P}(\%)$	$\mathcal{R}(\%)$	$\mathcal{F}_1(\%)$
1	93.06	89.74	97.22	93.33
2	61.81	62.32	59.72	60.99
3	96.53	100.00	93.06	96.40
4	75.00	86.00	59.72	70.49
5	61.11	76.67	31.94	45.10
6	65.97	62.92	77.78	69.57
7	80.56	86.67	72.22	78.79
8	95.83	93.42	98.61	95.95
9	93.06	96.97	88.89	92.75
Average	80.99	83.52	75.35	78.93

dataset or a different one. The intervariability criteria assesses whether the method accounts for differences between subjects by not relying on the same configuration across all of them. Notably, for the GLCSP method, as reported in [1], the dataset used was measured on six subjects rather than the typical nine. This discrepancy is significant when comparing it with other methods, such as the VPCSP [4], which used a different dataset consisting of ten subjects. Among the methods presented, the WTRCSP [7] and WOLA-CSP [5] variants show both a high reported accuracy (79.67% and 78.86%, respectively) and strong adaptability, suggesting their potential to be adapted to various real-world applications. In contrast, methods like Filter Bank CSP (FBCSP) [15] demonstrate lower accuracy (73%) and have limited adaptability, making them less versatile in comparison.

The most promising performance is observed in the XAI-CSP method, with an average classification accuracy of 80.99%, which indicates its effectiveness across different datasets and its potential for wider application.

In summary, while the dataset and adaptability factors vary across the methods, the XAI-CSP method emerges as the most robust, offering the highest accuracy and flexibility for future applications.

TABLE II: A Comparison of various CSP variants and their adaptability

CSP variant	Accuracy (%)	Dataset	Intervariability
GLCSP [1]	74.13	$\times$: 7	$\times$
VPCSP [4]	76.06	$\times$: 10	$\times$
WTRCSP [7]	79.67	✓	✓
FBCSP [15]	73	✓	$\times$
WOLA-CSP[5]	78.86	✓	✓
XAI-CSP	80.99	✓	✓

The integration of the RuleFit algorithm into the XAI framework significantly enhances interpretability and allows for dynamic feature selection. However, this integration also introduces additional computational overhead due to the generation of rules from DT. While this overhead is manageable in offline settings, it may pose challenges in real-time brain-computer interface BCI applications, particularly in resource-

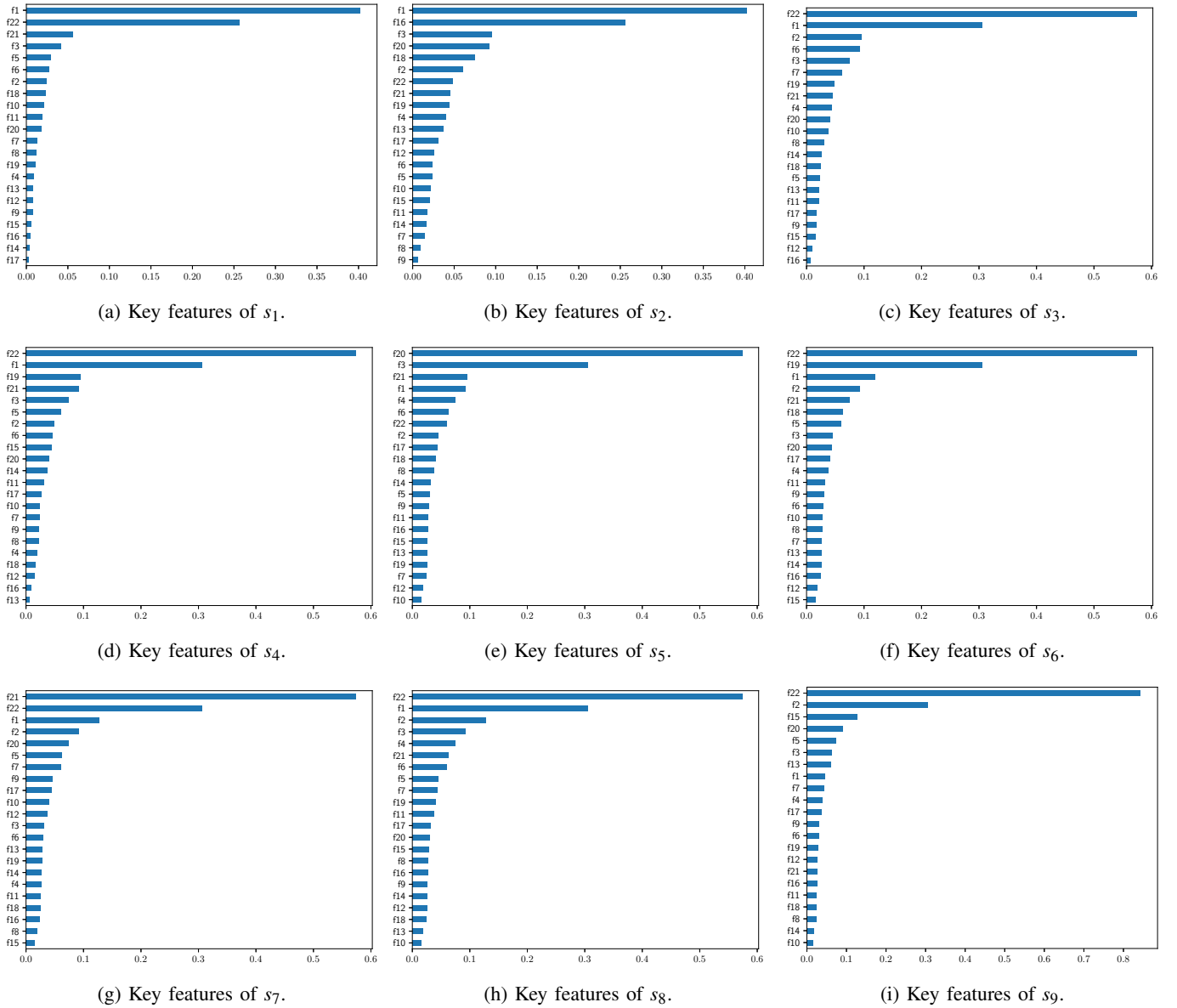


Fig. 2: Extracted features $\mathcal{W}$ sorted by the importance based on the RuleFit for different subjects.

constrained or embedded systems where latency and computational efficiency are critical. To address this issue, future research should explore model compression techniques or approximate rule inference to maintain interpretability while reducing computation time. Additionally, the applicability of the proposed XAI-CSP framework to real-time scenarios can be improved by pre-training the RuleFit component offline and then deploying only the most relevant subset of features and rules during inference. Integrating XAI into existing real-time BCI platforms will require optimizing the processing pipeline for low-latency execution. This could involve using graphical processing unit (GPU) acceleration, hardware acceleration [16], edge computing, or hybrid offline-online architectures. By balancing interpretability with efficiency, XAI has the potential to become a practical and explainable solution for

next-generation real-time BCI systems.

IV. CONCLUSION

In this study, we proposed an innovative framework integrating CSP with XAI to address key limitations in traditional CSP approaches for BCIs. The proposed method enhances classification performance by introducing explainability metrics and dynamic feature selection while offering valuable insights into the spatial patterns contributing to EEG-based MI tasks. This hybrid approach improves the interpretability and provides a flexible mechanism for adapting to inter-subject variability. Our results demonstrate that the Explainable CSP framework outperforms state-of-the-art CSP variants, achieving nearly 3% improvement in average classification accuracy compared to the basic CSP. Furthermore, the explainability component

offers a robust mechanism to dynamically optimize feature selection and provides practitioners with actionable insights for system design and deployment. This work represents a significant step towards more interpretable, adaptive, and reliable CSPs.

Future research directions include extending the proposed methodology to multiclass classification problems, enhancing the feature selection mechanism based on the feature's importance, incorporating additional datasets for validation, and exploring real-time implementation in wearable systems.

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